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Influence of plant and soil layer on energy balance and thermal performance of green roof system

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A B S T R A C T

This paper aims to make clear the effects of two important parameters, namely thickness of soil layer and leaf area index of plant layer, on green roof energy and thermal performance. Based on a coupled hygrothermal transfer model validated by field experiments in Shanghai area, energy balance of plant and soil layer is analyzed. Then 18 cases are simulated for different combinations of soil thickness and leaf area index, the results show that soil thickness has a significant effect on long term thermal performance of green roof in both summer and winter, while the effect of leaf area index is only great in summer. Compared with the hypothetical case without evaporation and transpiration, it is found that evaporation of soil layer makes a greater contribution in summer but a less one in winter compared with insulation effect of soil layer. For plant layer, shading is the main cooling way of roof deck while transpiration plays a minor role. Finally, based on the simulation results of long-term thermal performance, two new indexes are proposed to evaluate the relative thermal characteristic of green roof. And the influences of soil thickness, leaf area index and evapotranspiration on the two indexes are analyzed respectively.

1. Introduction

Green roof, which is also called as planted roof, has existed for a long time. The most famous example is the Hanging Gardens of Babylon, and it has been considered as one of Seven Wonders of the World in ancient times for its unique appearance and design. Nowadays, more and more attentions are paid on expanding the benefits of green roofs, such as mitigating heat island effect, reducing energy consumption of cooling etc. [1–3]. The government of Shanghai ruled that more than 30% of the available roof area should be covered by green roof for newly built public buildings after the year of 2015 [4]. Different from traditional roofing systems, green roof system is generally made up of plant layer, soil layer, filtering layer, drainage layer and structure layer [5]. Although the energy and thermal performance of green roof under different climates has been discussed in the past decades [6–12], there is not a consistent view about the contribution of each part to the total thermal performance of green roof, especially for plant layer and soil layer. Through a comparison test between earth-sheltered roof and green roof in Israel, Schweitzer [13] found that the cooling effect of earth-sheltered roof was significantly smaller than green roof, suggesting that plant layer plays an important role. The same conclusion was also drawn by Wong [14] through a measurement in Singapore. A field experiment carried out by Feng et al. [15] in Guangzhou indicated that a quasi-saturated medium layer resulted in 58.4% of net radiation being dissipated by evaporative latent heat and a higher water content ratio contributed to improve thermal performance of green roof. However, another experiment under hot climate of North-east Italy revealed that temperature differences between bare rooftop and green roof modules were higher at lower soil water content, hence Nardini et al. [16] deduced that plant transpiration had limited impact on green roof thermal performance. Sun [17] further noted that the contribution of evapotranspiration depended on local climate, and was mainly controlled by energy availability and water availability. For the effect of soil layer, the study by Jim [18] showed that soil temperature of an intensive green roof remained rather stable at 10, 50 and 90 cm depth, which suggested that thermal performance of green roof

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Nomenclature

C_p	Specific heat [J/K/kg]
d_f	Thickness of leaf [m]
D	Diffusion coefficient [m^2/s]
E	Latent heat flux [W/m^2]
H	Sensible heat flux [W/m^2]
h_i	Indoor convection coefficient [$W/m^2/K$]
h_f	height of plant canopy layer
K	Hydraulic conductivity [m/s]
L	Latent heat of vaporization [J/kg]
LAI	Leaf area index
q_{ci}	Absolute humidity of canopy layer [g/kg]
q_l, q_v	Liquid and gas water flux [$kg/m^2/s$]
q_h	Heat flux through soil layer [W/m^2]
R	Radiation [W/m^2]
S	Water absorption ratio of root [kg/s]
T	Temperature [°C]
P	Precipitation [m]

Greek symbols

θ	Water content ratio [m^3/m^3]
λ	Heat conductivity [W/m/K]
ρ	Density [kg/m^3]
φ	Insulation factor
β	Sol-air temperature regulation factor

Subscripts

a	Air
c	Canopy space
f	Foliage layer
g	Soil layer
i, j	Calculation node
LR	Long wave radiation
s	Structure layer of green roof
r	Reference height
SR	Solar radiation
w	Water

could be obtained with only 10 cm soil. On the contrary, in the experiment of Nardini et al. [16], significant difference in terms of rooftop temperature reductions was found between 120 mm and 200 mm soil thickness. Different from previous two opinions, Sun et al. [17] argued that the effect of soil thickness on green roof thermal performance was not linear. Although thick soil had a higher thermal resistance, it also could store more water at the bottom and limit the evaporation of soil surface. And there was an optimum value of soil thickness for green roof thermal performance. Roche [19] further pointed out that, depending on the local climate, both soil depth and leaf area index played different roles in affecting the peak cooling load. For example, leaf area index is the most significant factor in Los Angeles (heating and cooling climate), while the soil depth makes the most contribution in Chicago (heating dominated climate). And an investigation of green roof impact on energy saving in Toronto by Berardi [20] also indicated that the building energy consumption after retrofitting was found to be more influenced by the soil depth than the leaf area index of green roof, but the cooling demand was mainly affected by leaf area index.

From the reviewed literature, it could be found that the effects of plant and soil layer vary in different conditions and lead to diverse thermal performance, thus it is not reasonable to evaluate or treat green roof as traditional insulation materials. In order to predict thermal performance of green roof, many types of models have been developed based on different assumptions. He et al. [21] developed heat transfer model based on Bowen ratio of green roof, which did not consider heat transfer between plant and soil. Barrio [22] took into account the moisture transfer process of plants, however, this model treated the water content of soil as a constant. Sailor [7] proposed a model based on Army Corps of Engineers' FASST vegetation models and implemented it into EnergyPlus software. Ouldboukhitine et al. [23] put forward a thermodynamic model based on energy balance and water balance equations expressed for foliage and soil media, and coupled it to TRNSYS code. Alexandri et al. [9] developed a dynamic green roof model which took into account heat and moisture transfer in the air close to roof surface and the effect of vapor diffusion on heat transfer. The above models were all validated by field experiments, while Tabares-Velasco et al. [24,25] put forward a quasi-steady heat and moisture transfer model and validated it by a new 'Cold Plate' apparatus in an environmental chamber.

According to these models, the shading effect of plant, namely transfer of shortwave and longwave radiation is usually based on Beer's Law [22], and transpiration is mainly calculated dynamically based on empirical equations of vapor pressure difference and stomatal resistance [25]. Stomatal resistance is the measure of rate that moisture gets through stomata, which is directly affected by various environmental factors, such as outdoor temperature, solar radiation, water availability and water vapor pressure deficit etc. And the effects of these factors tend to interact with each other, which makes the total understanding of stomatal resistance hard to model. Based on experiments and observations, some researchers have proposed several functions to express stomatal resistance based on environmental factors or net photosynthetic rate, and nonlinear multiplicative approach based on Jarvis scheme is commonly used [25]. Other factors, such as plant type, also affect evaporation [26]. The effect of soil layer involves evaporation and insulation, and an accurate prediction of water content distribution is necessary. Based on the usual assumption of non-deformable porous medium with uniform properties, the moisture transfer in soil is often expressed as one-dimensional Richard model [21]. However, the soil layer for green roof is often very thin (no more than 15 cm for extensive green roof), and there are relatively higher temperature and water ratio gradients through the soil layer. Therefore, coupled heat and moisture transfer is more appropriate for water content prediction of green roof. In addition, the soil used in green roof is often lightweight substrate, and has larger water holding ability and permeability compared with the soil used in ground greening.

In summary, plant and soil layer are two of the most important components for green roof, and complicated physical processes are involved in the hygrothermal transfer of above two layers. In addition to the hygrothermal properties of themselves, the performance of plant and soil layer are also affected by local climate [27–29]. And the aim of this paper is to make clear the energy balance of both layers and make a quantitative analysis of the effects of above two layers on the seasonal thermal performance of extensive green roof under the climate of Shanghai area. Firstly, a coupled heat and moisture transfer model is presented and validated by field experiment data. Compared with previous models, the model not only considers coupled heat and moisture transfer in soil layer, but also incorporates the multiple reflection and transmission of solar and longwave radiation in canopy space. In addition, the model takes into account the aerodynamic resistance above and under the canopy layer following the famous SVAT model by Choudhury et al. [30]. Based on the green roof model, the energy balance for plant and soil layer are analyzed in different conditions. Then seasonal thermal performance of green roof under different combinations of soil thickness and leaf area index (hereinafter referred to as LAI) are predicted and evaluated respectively in summer and winter. Besides the impact of sensible heat reduction (including insulation effect of soil layer and shading effect of plant layer), the impact of latent heat (namely evaporation and transpiration of soil and plant) on green roof thermal performance are also analyzed. Finally, two indexes are proposed to evaluate green roof in terms of insulation, shading and evapotranspiration effect, and the influence of plant and soil layer on the two indexes are discussed. It's expected that the results obtained in this study could provide further reference for green roof design and evaluation in Shanghai area.

2. Model development and validation

The model developed in this paper is based on our previous study [31] which has considered the following three aspects: (a) multiple reflection of longwave and solar radiation in canopy layer of plant; (b) the effect of wind profile on turbulent transfer coefficient below and above canopy surface based on SVAT theory by Choudhury et al. [30]; (c) water absorption of root and coupled heat and moisture transfer in soil layer based on the theory of Philip & De Vries [32]; And commonly used assumptions are given as below:

(1) Green roof system is reduced into three major layers, including plant layer, soil layer and structure layer.
(2) Plant layer and soil layer are assumed to be horizontally homogeneous, and hygrothermal transfer happens only along the height direction.
(3) Ignoring water change of plant itself, and assuming that the water content variation of soil layer equals to the water loss through evapotranspiration.
(4) The plant canopy is assumed as a layer of semi-transparent medium, where radiation obeys Beer's Law.
(5) The structure layer is waterproof, and moisture transfer through this layer is not considered.
(6) The biochemical processes (such as photosynthesis and respiration) and heat transfer through plant stems are ignored, and air beneath stomatal pores is saturated with water vapor.

2.1. Brief introduction of green roof model

Heat and moisture transfer model of green roof is illustrated in Fig. 1(a), and the basic equations are as follows. The more detailed description could be seen in Ref. [22].

(1) Energy balance equation of plant canopy layer

The energy balance of canopy layer considers sensible and latent heat transfer between foliage and canopy air as well as multiple reflection of shortwave and longwave radiation in canopy space:

$$(\rho C_p)_f d_f \mathrm{LAI_i} \frac{dT_{fi}}{d\tau} = R_{SR,fi} + R_{LR,fi} + H_{fi} + E_{fi} \tag{1}$$

where $(\rho C_p)_f$ is volumetric heat capacity of plant layer, d_f is leaf thickness, T_{fi} is average temperature of each part of plant layer, τ is time, $R_{SR,fi}$ and $R_{LR,fi}$ are solar radiation and longwave radiation of plant layer, H_{fi} and E_{fi} are sensible heat and latent heat transfer of plant layer respectively.

(2) Energy and moisture balance equation of soil surface

In addition to sensible and latent heat transfer between soil surface and canopy air, the energy and moisture balance equations of soil surface also consider sensible and latent heat transfer from underlying soil.

$$-\lambda_g \frac{\partial T_g}{\partial z}\Big|_{z=0} - L\left(D_{v\theta}\frac{\partial \theta}{\partial z} + D_{vT}\frac{\partial T_g}{\partial z}\right) = R_g + H_g + E_g \tag{2}$$

$$\rho_w \frac{d\theta}{d\tau} = -\frac{\partial (q_l + q_v)}{\partial z} - \frac{E_g + S - P}{\Delta z \mathrm{L}} \tag{3}$$

where λ_g is thermal conductivity of soil layer, T_g is temperature of soil layer, L is latent heat of moisture, $D_{v\theta}$ and D_{vT} are transfer coefficient of water vapor under gradient of moisture and temperature, θ is volumetric water ratio of soil layer, R_g is net radiation of soil surface, H_g and E_g is sensible heat and latent heat transfer of soil surface, ρ_w is density of liquid water, Δz is grid step of soil layer, q_l and q_v are the mass flow of liquid water and gas water transfer, S and P are evaporation water loss and precipitation.

(3) Energy and moisture balance equation in canopy space

Energy and moisture balance of canopy air not only include energy and moisture transfer from plant leaves and soil surface, but also cover energy and moisture transfer with outside air at reference height.

$$(\rho C_p)_a \left(h - d_f \mathrm{LAI}\right) \frac{dT_c}{d\tau} = -H_f - H_g - H_a \tag{4}$$

$$\rho_a \left(h - d_f \mathrm{LAI}\right) \frac{dq_{ci}}{d\tau} = \frac{E_f}{\mathrm{L}} + \frac{E_g}{\mathrm{L}} + \frac{E_a}{L} \tag{5}$$

where h is the height of plant canopy layer, H_a and E_a is sensible and latent heat transfer from reference height to canopy space, q_{ci} is absolute humidity of canopy layer.

(4) Coupled heat and moisture transfer equations in soil layer

According to Philip & De Vries [32], the moisture transfer of soil is driven by the gradients of temperature and water content, and heat transfer of soil involves heat conduction, air and moisture flow and phase change. Water content and temperature of soil have various impact on transfer coefficients, which couples heat and water transfer.

$$\rho_w \frac{\partial \theta}{\partial \tau} = -\frac{\partial (q_l + q_v)}{\partial z} - \frac{S}{\Delta z} \tag{6}$$

$$(\rho C_p)_g \frac{\partial T_g}{\partial \tau} + (\rho C_p)_w q_l \frac{\partial T_g}{\partial z} = -\frac{\partial q_h}{\partial z} \tag{7}$$

$$q_l = -\left(D_{l\theta}\frac{\partial \theta}{\partial z} + D_{lT}\frac{\partial T_g}{\partial z}\right) - K \tag{8}$$

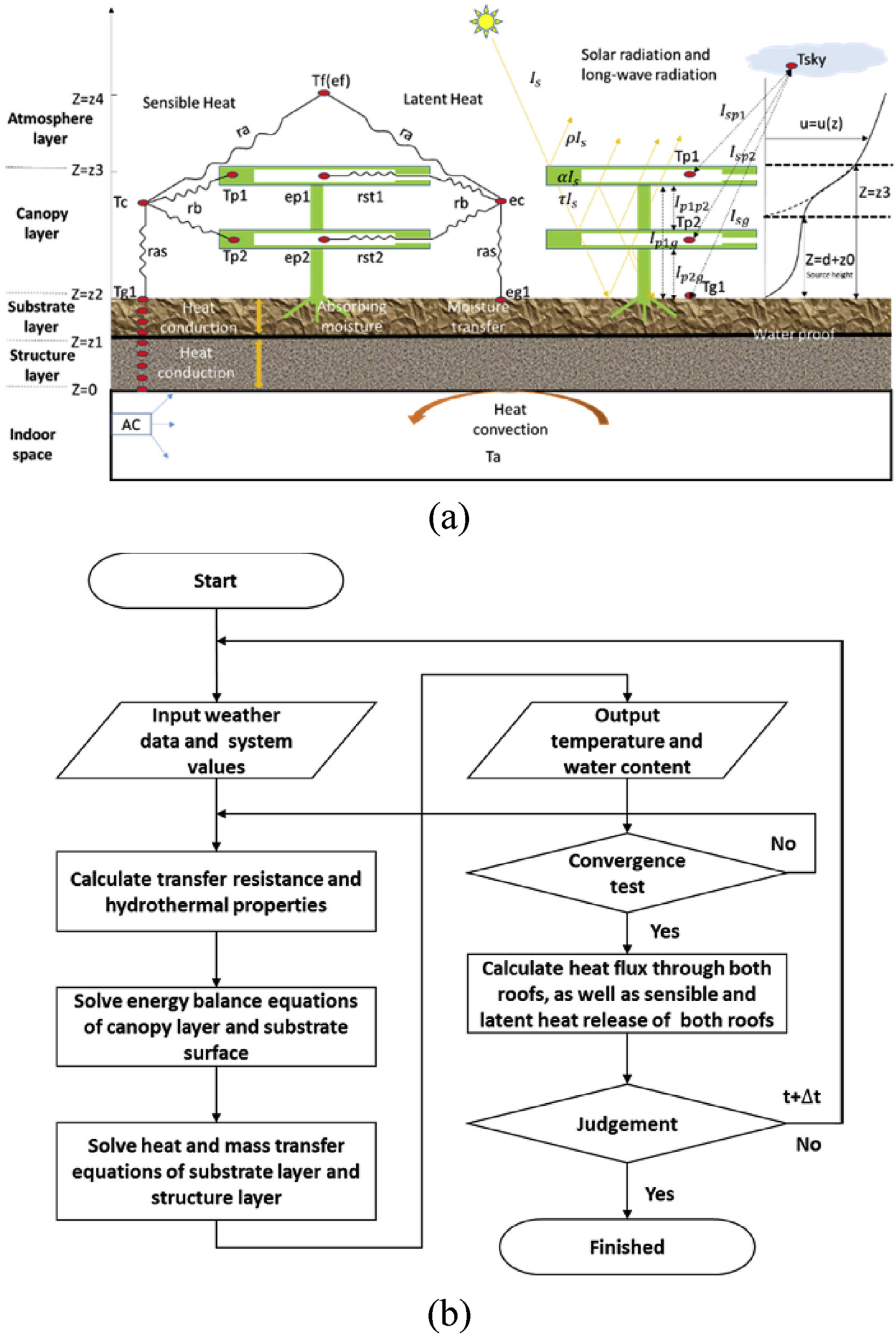


Fig. 1. (a) Schematic of heat and moisture transfer model for green roof; (b) flow-chart of modelling.

$$q_v = -\left(D_{v\theta}\frac{\partial\theta}{\partial z} + D_{vT}\frac{\partial T_g}{\partial z}\right) \tag{9}$$

$$q_h = -\lambda_g\frac{\partial T_g}{\partial z} - L\rho_w\left(D_{vT}\frac{\partial T_g}{\partial z} + D_{v\theta}\frac{\partial\theta}{\partial z}\right) \tag{10}$$

where $(\rho C_p)_g$ is volumetric heat capacity of soil layer, and q_h is the sum of sensible and latent heat transfer in soil layer, $D_{l\theta}$ and D_{lT} are liquid water transfer under the gradients of water ratio and temperature, K is water conductivity of soil layer.

(5) Heat transfer of structure layer

Compared with soil layer, the effect of moisture transfer in structure layer is minor, so only heat conduction is considered in this layer.

$$(\rho C_p)_s \frac{\partial T_s}{\partial \tau} = \lambda_s \frac{\partial^2 T_s}{\partial z^2} \tag{11}$$

where $(\rho C_p)_s$ is volumetric heat capacity of structure layer, T_s is temperature of structure layer, λ_s is thermal conductivity of structure layer.

(6) Convective heat transfer on the internal surface of structure layer

$$-\lambda_s \frac{\partial T_s}{\partial z}|_{z=0} = h_i (T_s|_{z=0} - T_{in}) \tag{12}$$

where h_i is indoor heat convection coefficient (8.7 W/m^2), T_{in} is indoor air temperature.

2.2. Model validation

In order to solve above equations, implicit finite difference scheme and predictor-corrector method are used to discretize and calculate those partial differential equations based on the programming software Matlab. The long-wave radiation and saturated vapor pressure are linearized by the same method performed by Sailor [7]. The timestep is 5 min. And after climate parameters and initial hygrothermal parameters are input, the program will calculate by successive iterations in each time step until convergence (as is illustrated in Fig. 1(b)).

A field experiment was carried out in Jiading Campus of Tongji University, and measurement data in summer and winter were recorded to validate and evaluate this model. As is shown in Fig. 2, the left roof was common roof that was covered by greenery modules, and the right one was just common roof that was made of foam sandwich panel. The prefabricated greenery modules (shown in Fig. 3(b)) combined plant layer, soil layer, filtering membrane and drainage layer together, and were connected with each other by buckles. The plant was sedum linear, and soil layer was comprised of peat soil, powdered perlite, vermiculite aggregate and organic fertilizer, which were both common in Shanghai area. During the experiment, the windows, doors of both room were locked, and indoor temperature was controlled by air conditioner at 26 ± 0.5 °C in summer and 24 ± 0.5 °C in winter. The detailed measurement plan is illustrated in Fig. 3(a), vertical temperature and moisture distributions as well as heat fluxes through green roof and common roof are all collected every five minutes, and detailed specifications of sensors are listed in Table 1. For green roof, the average temperatures of canopy layer and surface temperature of soil layer are two important parameters that reflect the energy balance of both layers, water ratio variation and heat flux into the room represent the intensity of evapotranspiration and thermal insulation performance of green roof respectively. Therefore, the measurement data of above four parameters are compared with the simulation results.

Fig. 2. Experimental setup of green roof and common roof.

The input data of above model are shown in Table 2, and weather data are presented in Fig. 4. A comparison between simulation results and measurement data is shown in Fig. 5, and it's obvious that green roof model is able to reproduce temperature and moisture variations with time. The values of RMSE and NSEC[1] [33] are calculated in Table 3 for above four parameters, which shows the model has a good accuracy and efficiency. It can be also found from this table that, the error of soil surface temperature in winter is greater than that in summer, while the error of soil water content in winter is smaller than that in summer. The reason for this result is that the withered vegetation formed an insulation layer above the soil layer in winter, which changed the energy balance of soil surface. And the intensity of evapotranspiration in cold winter was far smaller than that in hot summer, which reduces the prediction error of soil water content.

3. Energy balance analysis of plant and soil layer

To further reveal the difference between effects of plant and soil layer, energy balance of both layers on typical summer and winter days are presented in Fig. 6 based on above simulation results. In summer, net solar radiation and long wave radiation of plant layer is far greater than that of soil layer because of shading effect. Latent heat is the main heat dissipation way for both layers, which accounts for more than 50% of net solar radiation, and the other part of heat gain is dissipated by net longwave radiation and heat convection. It's also found that, due to higher temperature difference between plant layer and outside air, heat convection for plant layer is greater than that for soil layer. Heat conduction through soil layer in summer is very small and the average value is no more than 1% of average net solar radiation, which shows the excellent ability of green roof to resist heat transfer into the room. In winter, the LAI of plant layer is very small, and most of solar radiation is absorbed by soil surface. And for soil layer, longwave radiation and convective heat transfer become the main heat dissipation ways while the ratio of latent heat transfer is much smaller than that in summer due to lower solar radiation and outdoor temperature. Because of larger temperature difference between indoor space and outdoors, a lot of heat flows out of the room and the average heat flux is larger than that in summer. For heat storage of both layers, there are no significant difference in both seasons.

It has been proved that LAI of plant layer has a direct effect on

[1] RMSE refers to root-mean-square error, which is used for testing the reliability of a model; and NSEC refers to Nash-Sutcliffe efficiency coefficient, which is used for testing the efficiency of a model.

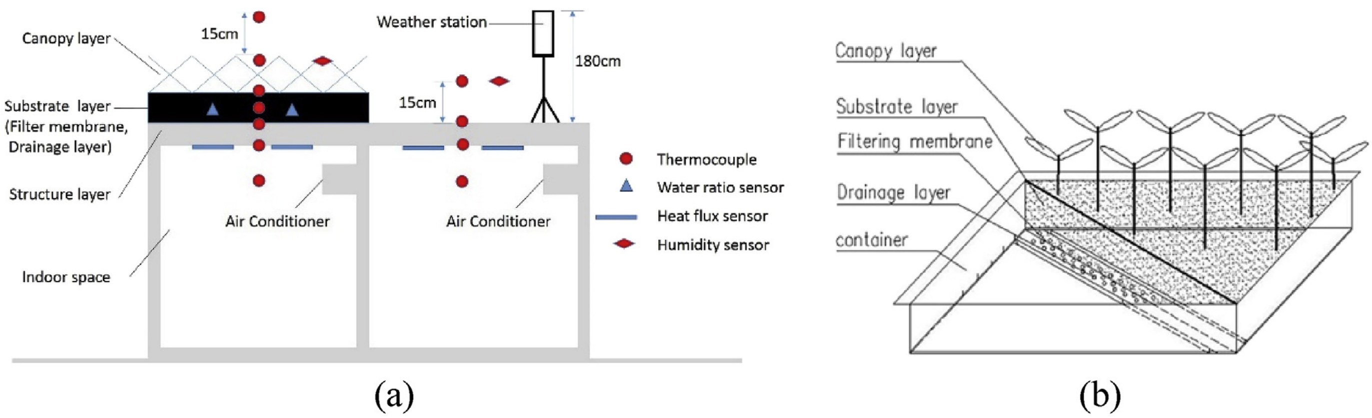


Fig. 3. (a) Schematic of experimental setup; (b) structure view of greenery module.

Table 1
Specifications of sensors in field measurement of green roof thermal performance.

Equipment	Type	Parameters	Resolution	Accuracy
Data acquisition meter	Agilent 34970A	–	–	–
Thermocouple	T	Temperatures of various layers	0.1 °C	±0.1 °C
Humidity Sensor	WSZY-1	Relative humidity	0.1%	±2%
Soil hygrometer	FDS-100	Water ratio of substrate	0.01%	±2%
Heat Flux meter	HFM-215N	Heat Flux	0.01 W/m^2	±3%
Weather station	TRM-ZS2	Temperature	0.1 °C	±0.2 °C
		Humidity ratio	0.1%	±3%
		Wind speed	0.1 m/s	±0.3 m/s
		Solar radiation	1 W/m^2	≤±5%
		Rainfall	0.1 mm	±4%

Table 2
Input data of green roof models.

Parameter	value	Parameter	Value
Height of plant (cm)	15(Summer), 6(Winter)	Thermal capacity of structure layer (J/m^{3}K)	1400*20
Minimum stomata resistance (s/m)	150	Thermal conductivity of structure layer (W/m·K)	0.04
Average LAI	5(summer), 0.5(winter)	Depth of structure layer (mm)	75
Soil thermal capacity (J/m^{3}K)	600*1000	Reflectivity of structure layer	0.2
Soil depth (cm)	4	Emissivity of plants	1
Soil conductivity (W/m·K)	$0.374*(\theta)^{0.403}$	Emissivity of structure layer	0.9
Reflectivity of leaves	0.3	Emissivity of soil surface	0.9
Soil water conductivity (m/s)	$2.37*10^{-5}*(\theta)^{24.42}$	Field water capacity of soil	0.71
Soil water capacity (pa^{-1})	$1.575*(\theta^{-4.35})$	Soil reflectivity	0.2

the shading and evapotranspiration of green roof [34], hence the energy balance of plant and soil layer under different LAI are presented in Fig. 7. It's shown that, as LAI rises, all the energy fluxes of plant layer improve, while the corresponding energy fluxes of soil layer decrease in both summer and winter. The reason is that a higher LAI could intercept more solar and sky longwave radiation and thus stimulate the plant layer to release more sensible and latent heat, hence the soil surface receives less solar and sky longwave radiation, and releases less latent and sensible heat. Compared with LAI of plant layer, the thickness of soil layer has no direct impact on the solar and longwave radiation of both layer, and it mainly affects the heat flux getting into the room and the outer surface temperature of structure layer [16].

4. The effect of plant and soil layer on thermal performance of green roof

Based on typical meteorological year data of Shanghai and the model developed above, seasonal thermal performance of green roof and corresponding common roof (thickness: 24 cm, average thermal conductivity: 1.51 W/m·K, average heat capacity: 2300000 J/m^{3}K) for a traditional non-insulated building in Shanghai area are simulated. As is shown in Table 4, there are 18 cases of green roof, which include three soil thickness and three LAI in summer and winter. Indoor temperature is set as 25 °C in summer and 18 °C in winter. The effect of soil thickness on thermal performance of green roof (GR) is shown in Fig. 8 (a) and (b), and thermal performance of common roof (CR) is also presented at the same time. Fig. 8 shows that, when soil thickness is larger, the fluctuation range of heat flux through roof deck is smaller and the delay time of heat flux wave is longer. Moreover, the heat flux changes little when thickness is up to 15 cm, which is consistent with the viewpoint of Jim et al. [18]. Owing to the little average temperature difference between indoor and outdoor (the average outdoor temperature is 25.59 °C), it's easy to understand that, even when soil thickness is larger than 15 cm, insulation performance

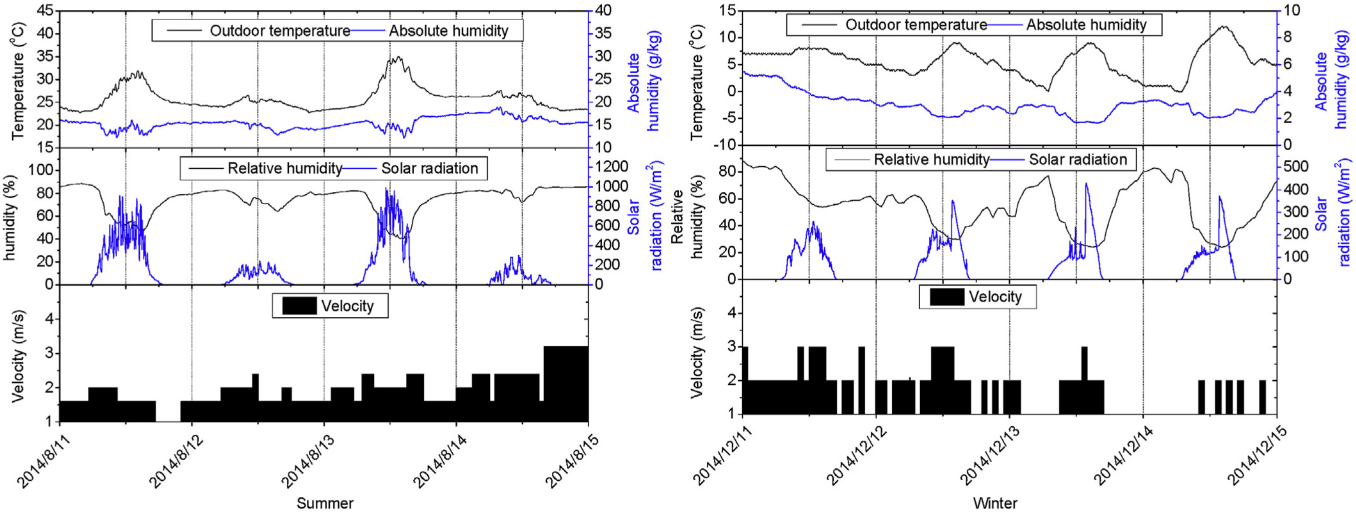


Fig. 4. Meteorological data during experiment on typical summer days and winter days.

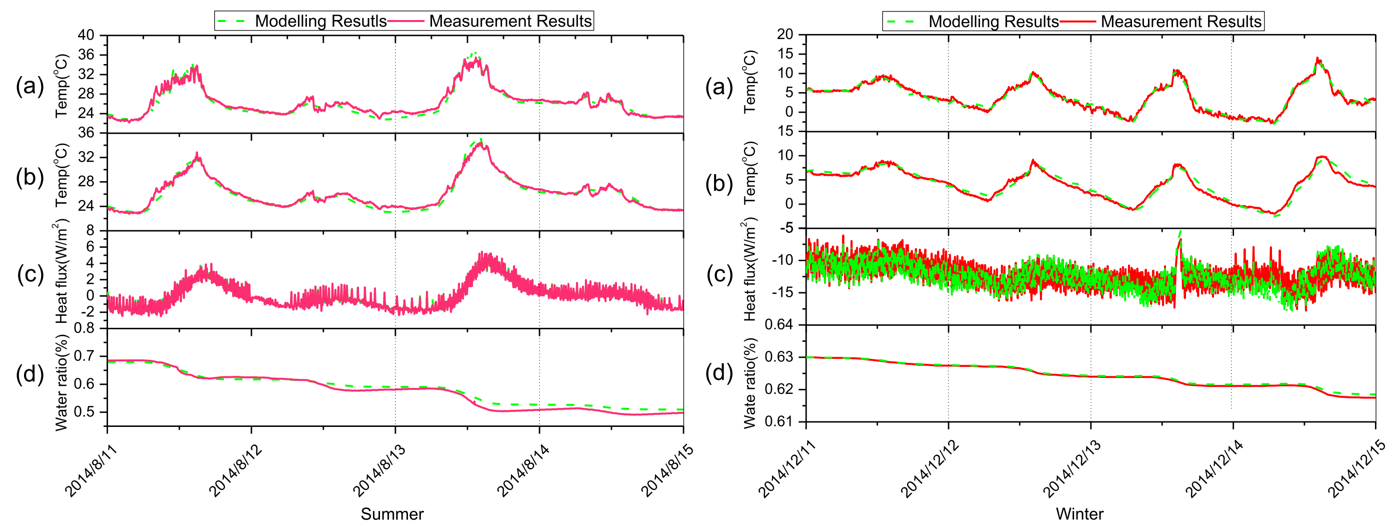


Fig. 5. Model validation of green roof in summer and winter: (a) average foliage temperature of canopy layer; (b) outer surface temperature of soil layer; (c) heat flux through green roof; (d) average water ratio of soil layer.

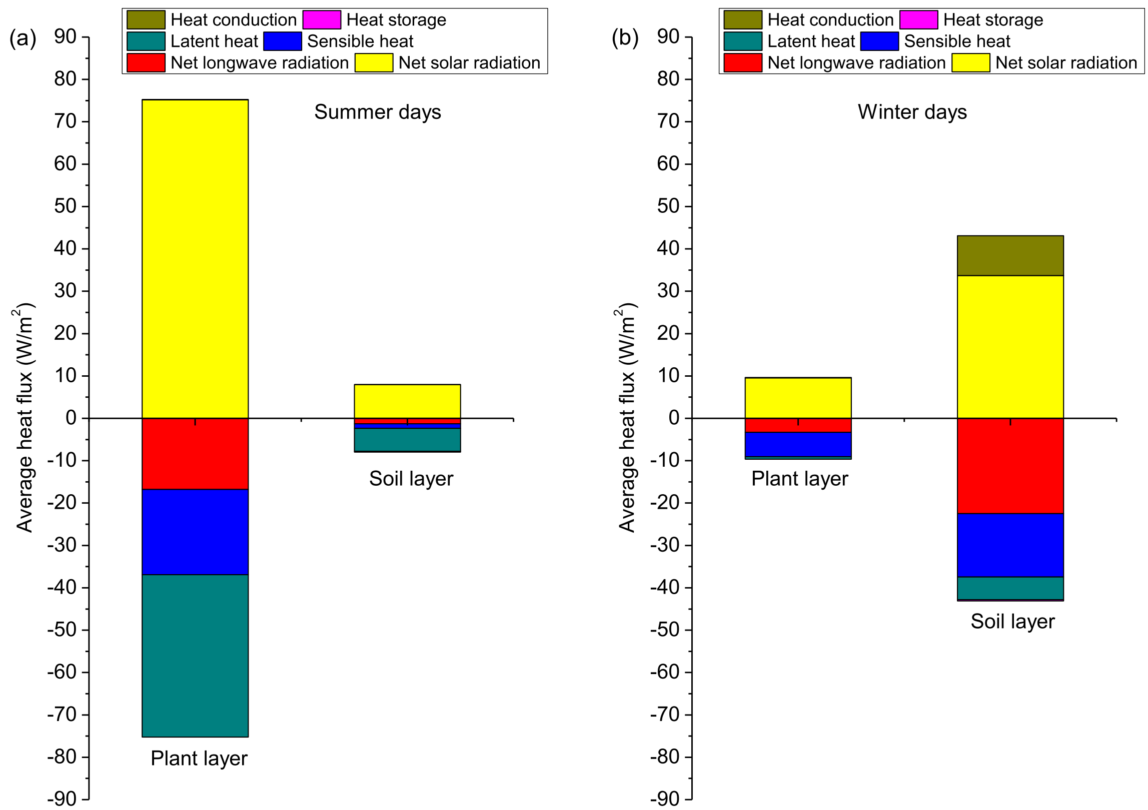
Table 3
RMSE and NSEC values of simulation parameters of green roof.

Parameters	Summer		Winter	
	RMSE	NSEC	RMSE	NSEC
Average temperature Of plant canopy (°C)	0.76	0.94	0.71	0.96
Temperature of Soil surface (°C)	0.52	0.96	0.69	0.95
Heat flux through green roof (W/m^2)	0.22	0.98	0.88	0.85
Average water content of soil layer (m^3/m^3)	0.01	0.94	0.00057	0.98

would not increase significantly in summer. However, in winter, because of the great temperature difference between indoor and outdoor, improving soil thickness could evidently decrease heat flux through roof deck. The impact of LAI is illustrated in Fig. 8 (c) and (d), a higher LAI reduces solar radiation reaching soil surface, and leads to a smaller fluctuation range and value of heat flux. Different from the impact of soil thickness, LAI does not influence the delay time of heat flux wave. In winter, because of less solar radiation and lower LAI, the heat flux variation is less affected by LAI. The seasonal average heat flux through roof deck for all the 18 cases are presented in Fig. 9, it shows that no matter in summer and winter, soil thickness has a significant impact. However, LAI only has a large effect in summer and does not show obvious effect in winter. Besides, as soil thickness and LAI are larger, the influence of both factors on green roof thermal performance gradually become small.

5. The effect of evaporation effect of soil layer on thermal performance of green roof

On typical summer days, insulation effect and evaporation effect both contribute to reduce heat flux through roof deck for green roof; on typical winter days, insulation effect benefits to decrease heat flux while evaporation increases heat flux. Therefore, the contribution of evaporation effect on green roof thermal performance is different for summer and winter. To quantify this effect, the cases in Table 4 are simulated without considering evaporation effect, and the results are compared with original cases. As is



Summer	net solar radiation	net longwave radiation	sensible heat	latent heat	heat storage	heat conduction
plant layer	75.20094	-16.78	-20.11	-38.34	0.031	--
soil layer	7.97789	-1.28	-1.08	-5.387	-0.242	0.0138
Winter	net solar radiation	net longwave radiation	sensible heat	latent heat	heat storage	heat conduction
plant layer	9.55344	-3.29	-5.74	-0.612	0.0933	--
soil layer	33.70177	-22.48	-14.97	-5.38	-0.274	9.397

Fig. 6. Energy balance for plant layer and soil layer of green roof system: (a) summer case; (b) winter case.

illustrated in Fig. 10, due to stronger solar radiation and higher outside air temperature, the intensity of evaporation effect is much greater in summer, thus the impact of evaporation on reducing heat flux through roof is larger. In addition, as soil thickness is larger, thermal resistance of soil layer becomes higher, which leads to less impact of evaporation on heat flux. And as LAI is higher, less solar radiation reaches soil surface and results in lower surface temperature and evaporation intensity, thus the impact of evaporation on heat flux decreases. It's also shown that, the contribution of insulation effect has no obvious difference compared with that of evaporation effect in summer but plays more important role in winter.

6. The effect of transpiration effect of plant layer on thermal performance of green roof

In contrast with soil layer, plant layer affects thermal performance of green roof in another two ways, namely shading and transpiration effect, and these effects greatly depend on LAI. Improving LAI results in stronger solar shading effect and transpiration rate, which could reduce the soil surface temperature. In order to distinguish the difference between these two effects, the cases in Table 4 are simulated without considering transpiration effect, and a comparison with original cases are presented in Fig. 11. It demonstrates that transpiration has little impact on heat flux through green roof, which is consistent with the findings of Nardini et al. [16]. The main reason lies in the fact that transpiration mainly affects the average temperature of plant canopy layer, and thus affects soil surface temperature through long wave radiation indirectly. From the comparison results, it can be concluded that plant canopy layer affects green roof thermal performance mainly through shading effect, and transpiration effect only plays a minor role.

7. Evaluation of green roof long term thermal performance based on two new indexes

According to above analysis, it is easy to understand that green roof improves thermal performance of corresponding common roof mainly through insulation, shading and evapotranspiration effect. In previous studies, most of indexes are proposed to evaluate thermal performance of green roof based on the average heat flux through green roof [13,35], or equivalent thermal resistance [36,37] and overall thermal transfer value (OTTV) [38] et al. Although these indexes are very important to understand the relative benefits of green roof, they cannot reflect the internal mechanism. Under this condition, two new indexes are proposed here based on above three effects. As is illustrated in Fig. 12, average heat flux through green roof and common roof versus average indoor temperature in summer are presented. Based on equations (13) and (14), it is easy

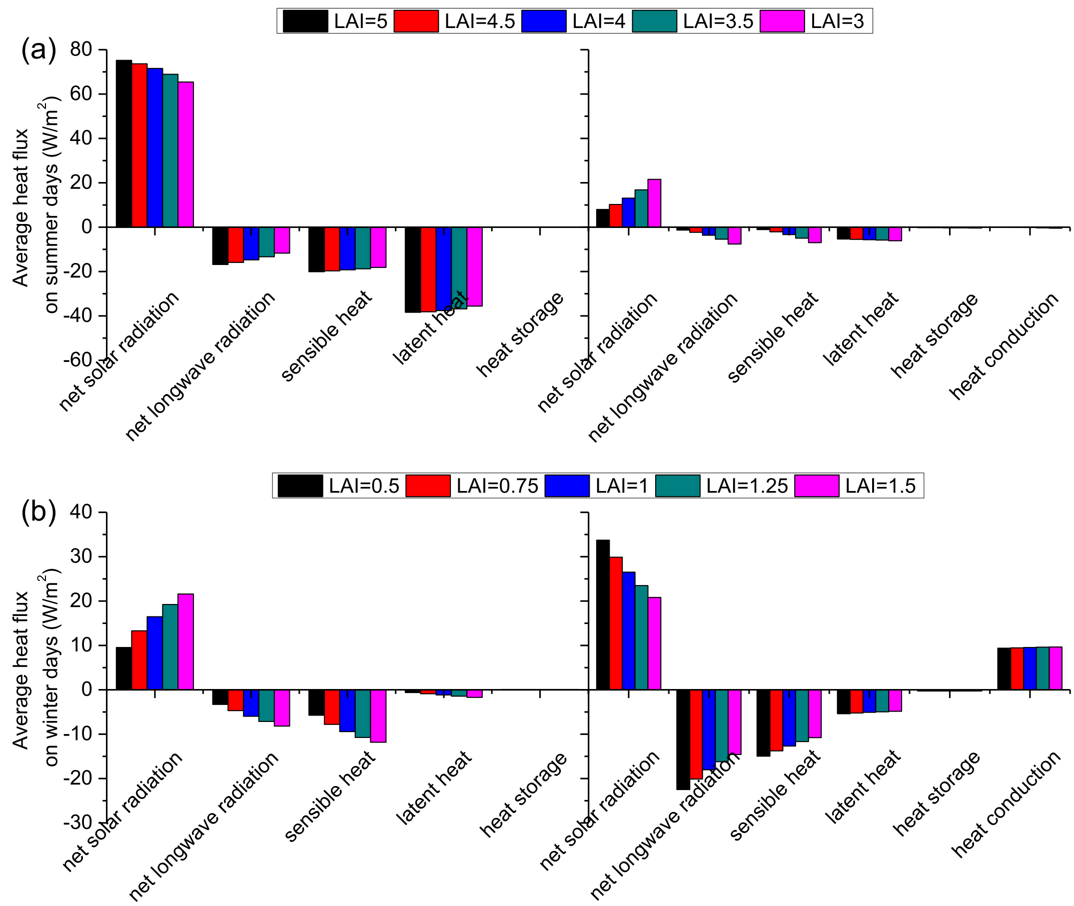


Fig. 7. The impact of leaf area index on energy balance of plant and soil layer: (a) Summer case; (b) Winter case.

Table 4
Cases for analyzing impact of soil and plant on thermal performance of green roof.

	Soil Thickness		
	4 cm	10 cm	15 cm
Summer Case			
LAI = 2	Case 1	Case 2	Case3
LAI = 3.5	Case4	Case5	Case6
LAI = 5	Case7	Case8	Case9
Winter Case			
LAI = 0	Case10	Case11	Case12
LAI = 0.5	Case13	Case14	Case15
LAI = 1	Case16	Case17	Case18

to know that the line slope in Fig. 12 represents the negative reciprocal of total thermal resistance, and X-intercept in Fig. 12 expresses average sol-air temperature[2] of outdoors. From Fig. 12, it's shown that the line slope and X-intercept of green roof are lower than that of common roof, which means that the total resistance of green roof is larger than that of common roof, and sol-air temperature of outdoors for green roof is lower than that for common roof. Therefore, as is shown in equations (15) and (16), two new indexes are proposed. The first index is called insulation factor which reflects the relative insulation effect of green roof, and the second one is called sol-air temperature regulation factor which reflects the total effect of insulation, shading and evapotranspiration on sol-air temperature of outdoors. According to above definition, these two indexes are calculated for summer and winter condition based on typical meteorological year data of Shanghai. The results are presented in Table 5. Obviously, φ has a variation less than 2%, which demonstrates that insulation effect of green roof changes little around the year. While β has a great variation, and the value in summer is obviously larger than that in winter. This indicates that green roof has a larger cooling effect in summer than that in winter, which is in consistent with the growth of plant.

$$R_t = \frac{\overline{T_z} - \overline{T_{in}}}{\overline{q}} = R_{out} + R_s + R_{in} \tag{13}$$

$$\overline{q} = -\frac{1}{R_t}\overline{T_{in}} + \frac{1}{R_t}\overline{T_z} \tag{14}$$

where, R_t is total thermal resistance, $\overline{T_z}$ is average value for sol-air temperature of outdoors, $\overline{T_{in}}$ is average indoor temperature, $\overline{q}$ is heat flux through roof, R_{out} is thermal resistance of external surface convection, R_{in} is thermal resistance of internal surface convection, and R_s is thermal resistance of roof.

$$\varphi = \frac{R_{t,g} - R_{in} - R_{out}}{R_{t,c} - R_{in} - R_{out}} \tag{15}$$

[2] Here sol-air temperature is a hypothetical temperature, it not only considers solar radiation and sky longwave radiation, but also takes into account the evapotranspiration effect for green roof, which reflects the comprehensive surface characteristic of both roofs.

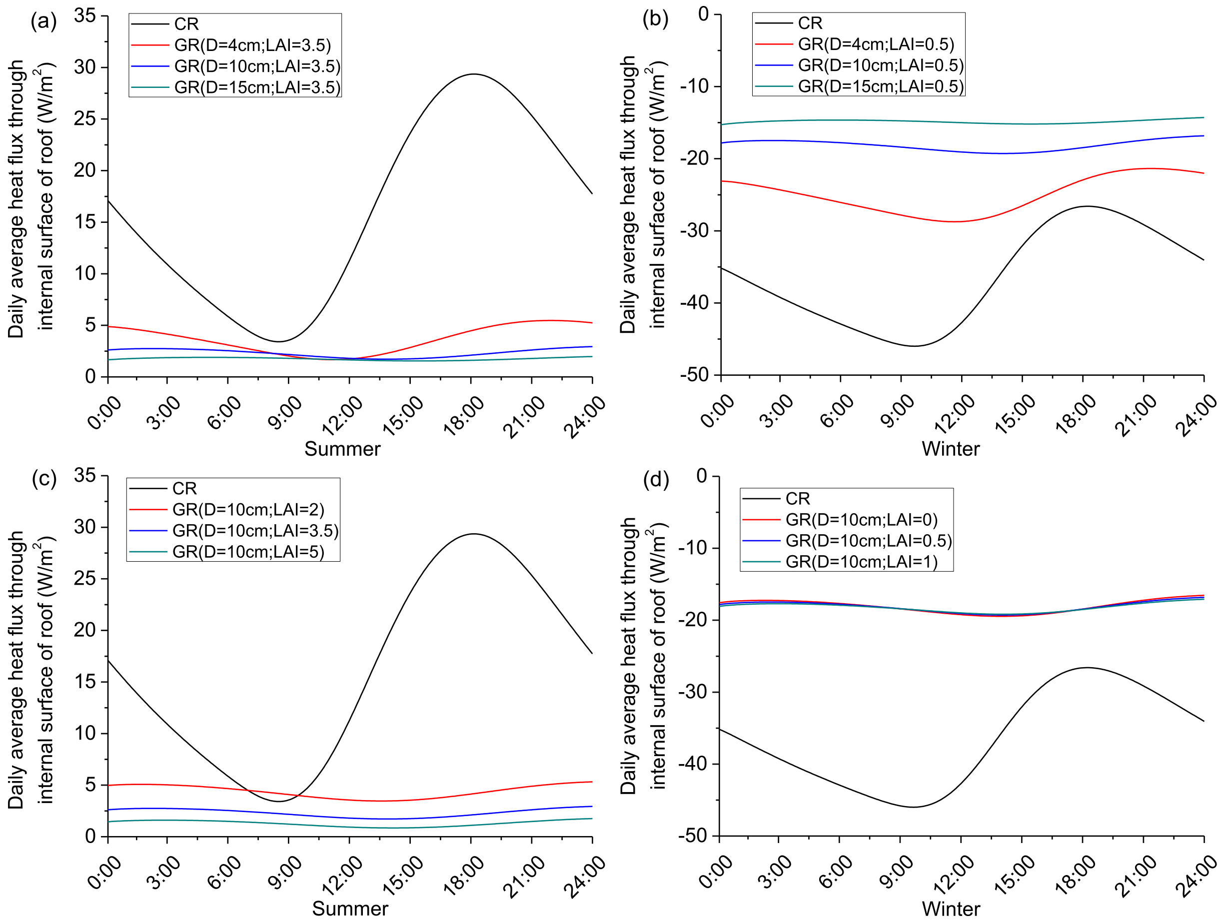


Fig. 8. The effect of soil thickness and LAI on thermal performance of green roof: (a) soil thickness effect in summer; (b) soil thickness effect in winter; (c) LAI effect in summer; (d) LAI effect in winter.

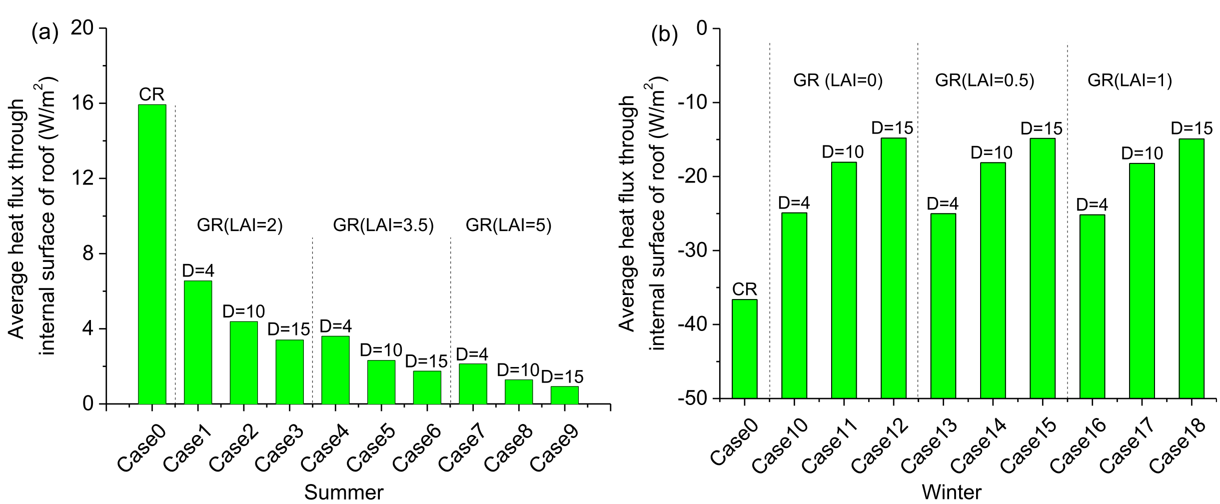


Fig. 9. Average outer surface temperature of roof deck for green roof under different combinations of soil thickness and LAI: (a) summer cases; (b) winter cases.

$$\beta = \frac{T_{z,c} - T_{z,g}}{T_{z,c} - T_{out}} \tag{16}$$

where, $R_{t,g}$ is total thermal resistance for green roof, $R_{t,c}$ is total thermal resistance of CR, $T_{z,c}$ is average sol-air temperature for CR, $T_{z,g}$ is average sol-air temperature for GR, T_{out} is average outdoor temperature.

Based on above analysis, it could be known that a larger β in summer and a smaller β in winter would contribute to better thermal performance of green roof. And a larger φ benefits to green roof thermal performance in both seasons. Previous sections have

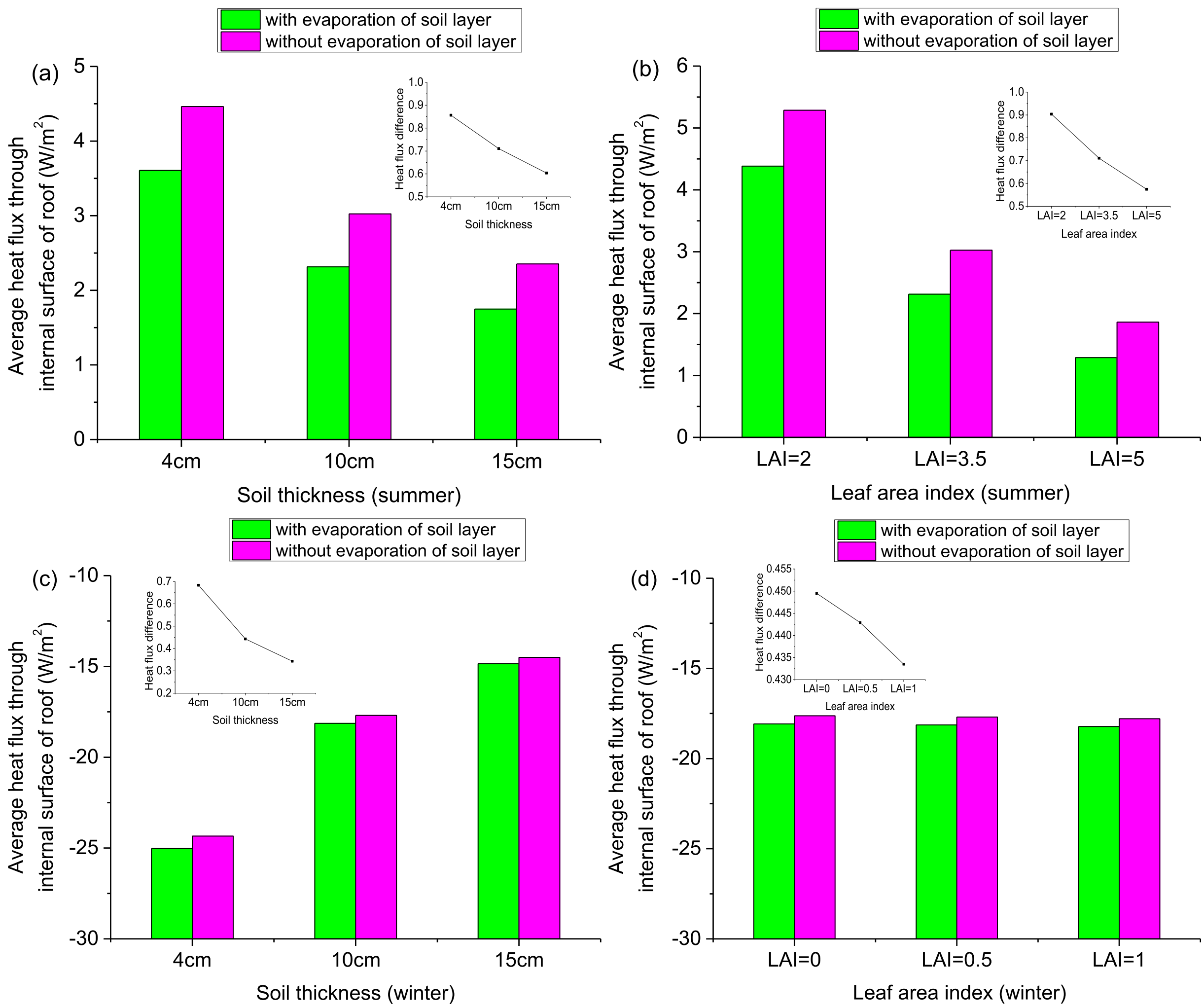


Fig. 10. The effect of soil evaporation on thermal performance of green roof: (a) the case of different soil thicknesses in summer; (b) the case of different leaf area indexes in summer; (c) the case of different soil thicknesses in winter; (d) the case of different leaf area indexes in winter.

demonstrated that green roof shows different thermal performance under various combinations of soil thickness and LAI. In order to analyze effects of the two factors on φ and β, above indexes are calculated for all the cases in Table 4. The results are presented in Fig. 13, it is shown that no matter in summer and winter, φ is mainly affected by soil thickness and has little relations with LAI. For β, the two factors have positive correlations in both seasons. And LAI is the main influence factor in summer, while soil thickness has a larger effect on β in winter. The main reason lies in the fact that green roof has a large LAI in summer, thus only a small part of solar radiation reaches the soil surface, and the corresponding common roof has a much higher average sol-air temperature (4.38 °C higher than average outdoor temperature), hence shading effect of plant on β is strong while the effect of soil thickness on β is weak. However, most part of plant canopy layer is dormant in winter, which leads to a large portion of solar radiation reaching the soil surface directly. And common roof has just a little higher average sol-air temperature (only 0.42 °C higher than average outdoor temperature), hence shading effect of plant is small while the effect of soil thickness on β is relatively strong. Evaporation and transpiration can reduce the average temperature of soil surface and plant canopy layer respectively, and φ and β are also calculated for the cases without evaporation and transpiration. From Fig. 14, it is shown that evaporation and transpiration almost have no obvious effect on φ but have a significant positive correlation with β in both summer and winter, and evaporation has a larger effect on β than transpiration. Based on the values of φ and β in this study, it is not only able to evaluate the relative thermal characteristic of green roof compared with corresponding common roof, but also able to predict the average long term thermal performance under different combinations of design parameters.

8. Discussion

Thermal performance of green roof is subjected to various factors, and plant density and soil thickness are only two important factors of them. It's known that water ratio has influence on thermal performance of plant and soil. On one hand, a higher water ratio helps to increase the evapotranspiration intensity; on the other hand, it decreases the thermal resistance of soil layer. Irrigation is the main way to control the water ratio of soil layer, and some studies have shown that the effect of irrigation on thermal performance of green roof depends on local climate conditions [17,39]. Shanghai belongs to subtropical monsoon climate with rich precipitation (more than 1100 mm on average throughout the year), and it is worthy to further make clear the effect of irrigation,

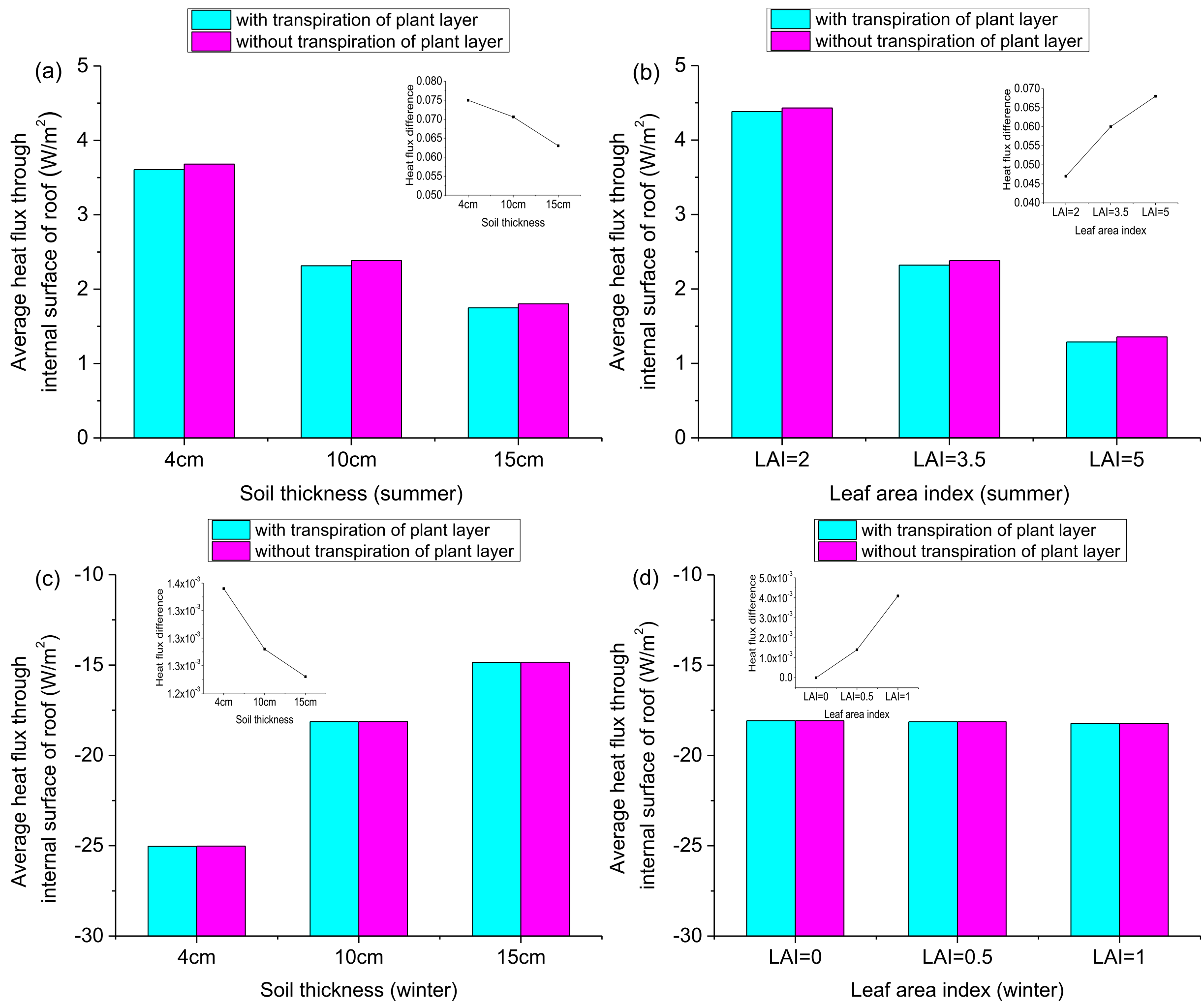


Fig. 11. The effect of plant transpiration on thermal performance of green roof: (a) the case of different soil thickness in summer; (b) the case of different leaf area index in summer; (c) the case of different soil thickness in winter; (d) the case of different leaf area index in winter.

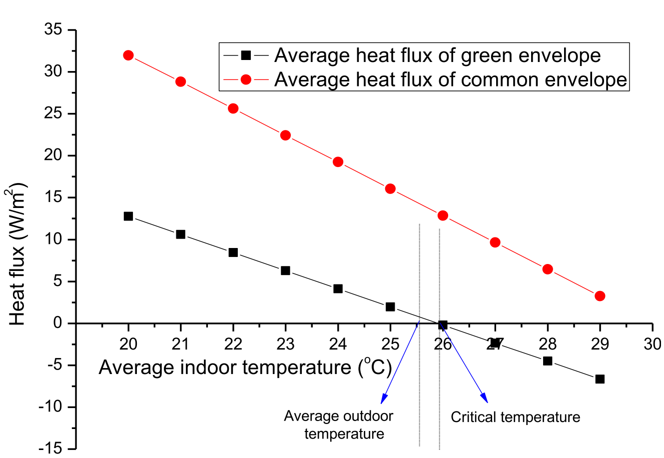


Fig. 12. Average heat flux variation through green roof and common roof versus average indoor temperature in summer.

especially smart irrigation, in terms of water-saving and thermal performance. In addition, some recent studies are focused on the thermal performance of roof systems that combine green roof with other components in order to make full use of the passive cooling and insulation effect of green roof. For example, Yeom [40] connected green roof with a radiant system that embedded the pipes in the soil layer, and Berardi [41] combined green roof with a water-to-air exchanger. Both systems have achieved better thermal performance than corresponding conventional green roof by enhancing the evaporative cooling effect. However, the effect of different types of evaporative cooling greatly depends on the local climate, and it is pretty necessary to carry out more experimental validation or numerical analysis for buildings with diverse types of green roofs and thermal properties under different climates. As to the model developed in this paper, more improvements are needed in the future. For example, the model only considers water balance of soil layer without taking into account rain interception effect of plant canopy layer and the surface runoff effect, which may cause bigger error on rainy days and the prediction of water ratio. And other factors, such as the thermal resistance of corresponding common roof and reflectivity of soil surface and foliage, also have influence on insulation factor and sol-air temperature regulation factor. It is valuable to build a database of above two indexes in the future and provide reference for design and evaluation of green roof system.

Table 5
Insulation factor and sol-air temperature regulation factor of green roof for summer and winter.

Season	Average outdoor temperature (°C)	Sol-air temperature (°C)		R (m^2 K/W)		φ	β
		$T_{z,g}$	$T_{z,c}$	$R_{t,g}$	$R_{t,c}$		
Summer	25.59	25.94	29.97	0.4629	0.3120	1.950	0.919
Winter	6.079	6.3432	6.5012	0.4654	0.3141	1.951	0.374

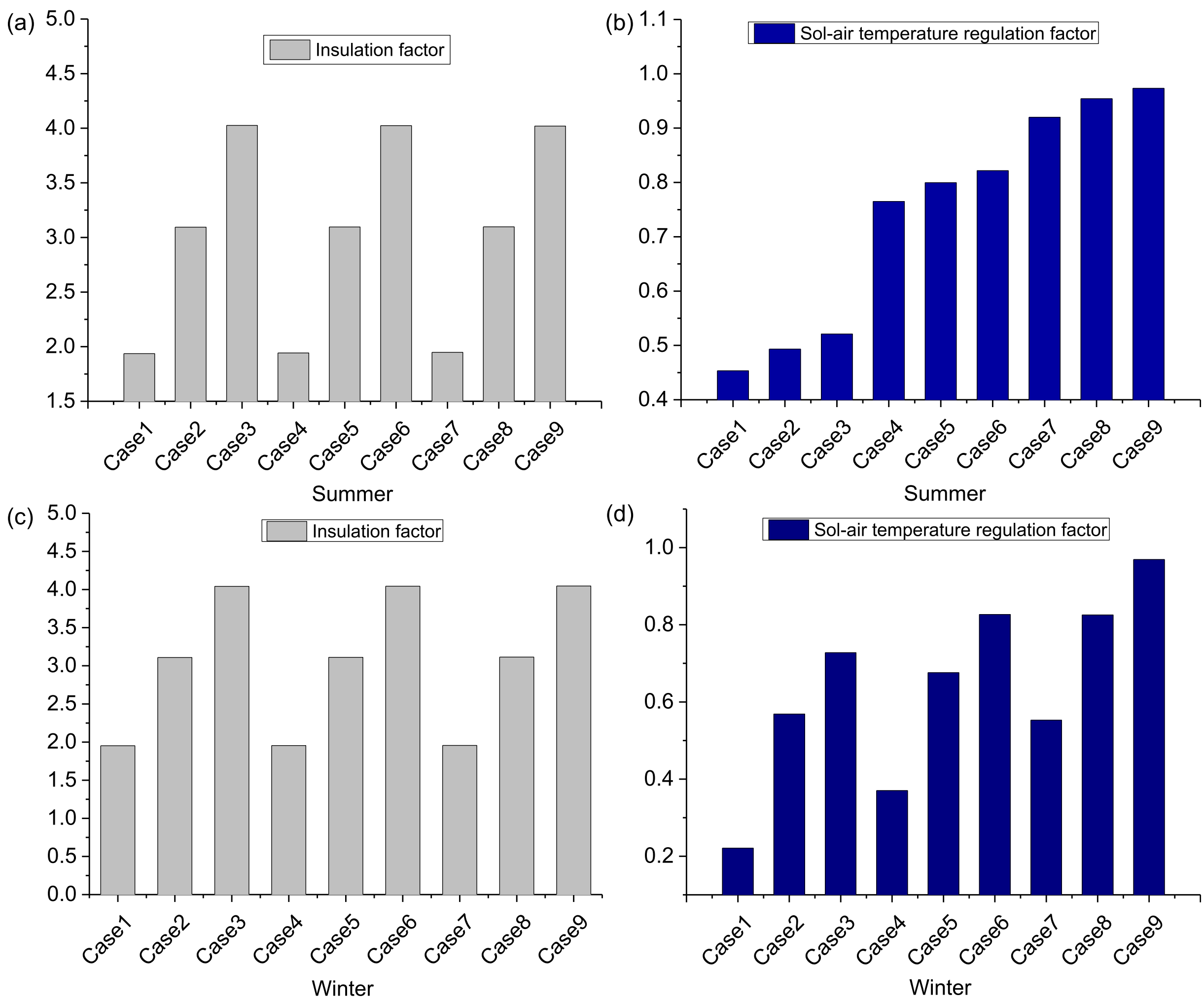


Fig. 13. Insulation factor and sol-air temperature regulation factor for different combinations of soil thickness and LAI in summer and winter: (a) insulation factor in summer; (b) sol-air temperature regulation factor in summer; (c) insulation factor in winter; (d) sol-air temperature regulation factor in winter.

9. Conclusions

Plant and soil play important role in green roof system, and this paper presents a detailed analysis of these two layers in terms of energy balance and long term thermal performance evaluation under the climate of Shanghai area. A coupled heat and moisture transfer model is firstly developed and validated by the field measurement data in summer and winter. Based on the simulation results of this model, the following conclusions can be drawn: (1) Net solar radiation is the main heat gain for both plant layer and soil layer in summer and winter, and latent heat is the main heat dissipation way for both layers in summer while longwave radiation is the main heat dissipation way in winter. Sensible heat dissipation of plant layer is far greater than that of soil layer in summer, but smaller in winter; (2) As leaf area index rises, all the energy fluxes of plant layer increase while corresponding energy fluxes of soil reduce; (3) Soil thickness has a noticeable effect on heat flux through green roof in both summer and winter, and the effect of LAI is only strong in summer. In addition, soil thickness also has positive effect on the delay time of heat flux wave while LAI has no obvious influence. As soil thickness and LAI increase, the effects of both factors become weak; (4) Compared with the insulation effect of soil layer, the effect of evaporation on green roof thermal performance is more significant in summer, and this effect becomes weak when LAI increases. In addition, transpiration almost has little effect on roof deck temperature compared with evaporation effect; (5) Two new indexes called insulation factor and sol-air temperature regulation factor are proposed in terms of insulation, shading and evapotranspiration effect. Soil thickness shows positive effect on these two factors while LAI only affects sol-air temperature regulation factor. In addition, evaporation and transpiration both have positive impact on sol-air temperature regulation factor, and

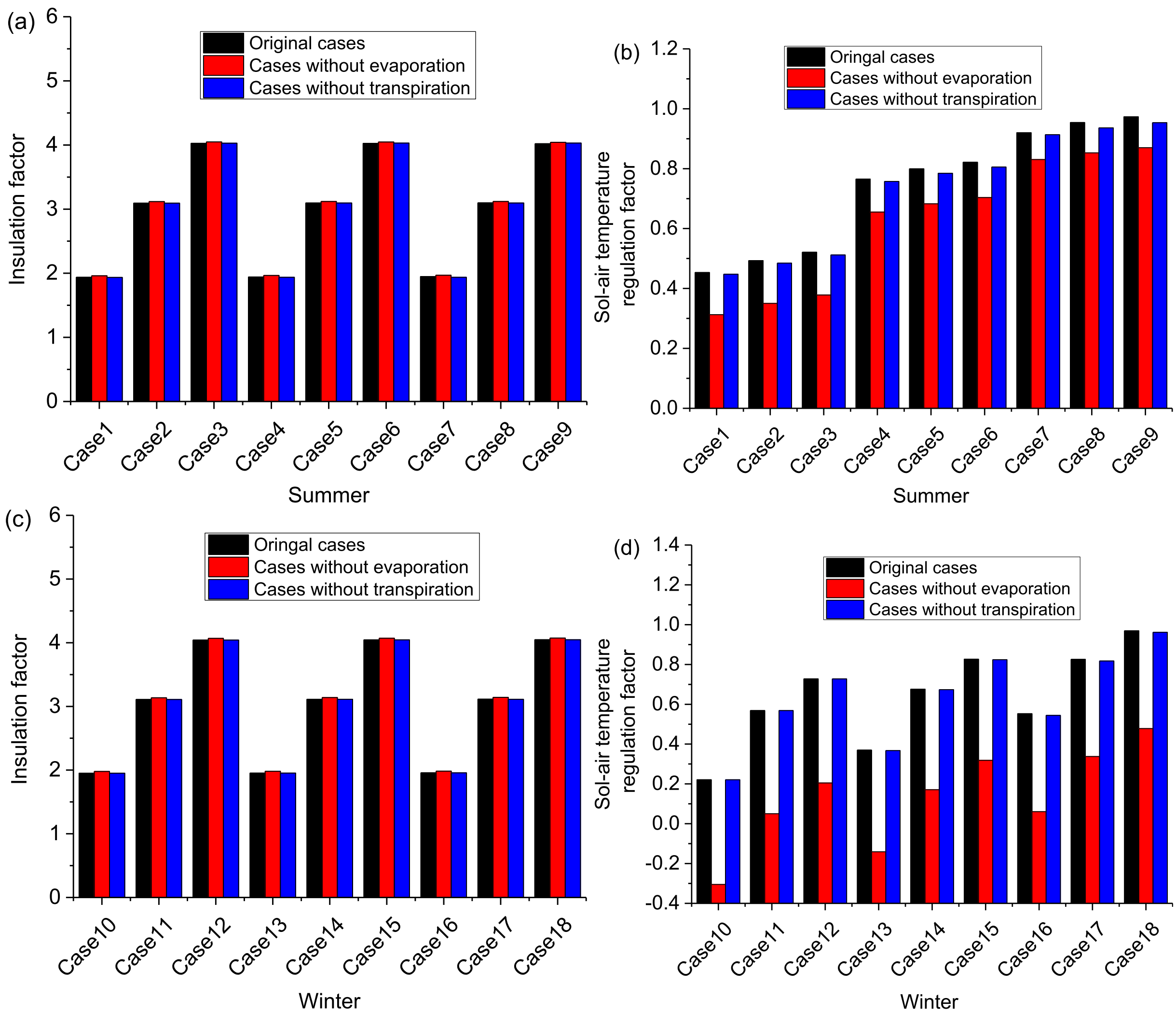


Fig. 14. Insulation factor and sol-air temperature regulation factor of cases with and without considering evaporation and transpiration effect in summer and winter: (a) insulation factor in summer; (b) sol-air temperature regulation factor in winter; (c) insulation factor in winter; (d) sol-air temperature regulation factor in winter.

the effect of evaporation is greater than that of transpiration. However, they both don't show clear effect on insulation factor.

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